Algorithm: Hospital Patient Management System

Main

Step 1: START. Create an empty doubly linked list, an array of 10 free beds and an empty stack.
Step 2: Display the menu: Admit, Discharge, Search, Display forward, Display backward, Bed status, Undo discharge, Discharge history, Exit.
Step 3: Read the choice and call the matching function.
Step 4: Repeat Steps 2 and 3 until choice = 0. STOP.

Admit Patient

Step 5: Find the first free bed in the array. If none is free, display "No beds available" and stop.
Step 6: Read patient ID. If the ID already exists in the list, stop.
Step 7: Read name, age and disease.
Step 8: Create a new node and insert it at the end of the list (update tail and prev/next links).
Step 9: Mark the bed as occupied and increase count.

Discharge Patient

Step 10: Read patient ID and search the list from head. If not found, display "Patient not found" and stop.
Step 11: Push the patient record onto the stack (top = top + 1).
Step 12: Free the patient's bed in the array.
Step 13: Delete the node by linking its previous and next nodes (update head or tail if needed).
Step 14: Decrease count.

Search Patient

Step 15: Read the patient ID or name.
Step 16: Traverse the list from head using next until NULL.
Step 17: If a node matches, display the patient details; otherwise display "Patient not found".

Display Patients

Step 18: Forward: start at head and print each node using next until NULL.
Step 19: Backward: start at tail and print each node using prev until NULL.

Undo Last Discharge

Step 20: If the stack is empty, display a message and stop.
Step 21: Pop the top record (top = top - 1).
Step 22: Insert the patient at the end of the list and mark a free bed as occupied.

Bed Status and Discharge History

Step 23: Bed status: traverse the bed array and print each bed as Free or Occupied.
Step 24: History: print the stack records from top to 0 (latest discharge first).